Pion Decay Rates in a 100 cm Tracking Volume

Kinetic Energy Impact on Tracking Efficiency

1. The Decay Formula

In the lab frame, the mean decay length λ is dilated by relativity:

$$\lambda = \beta\gamma c\tau$$

Where:

- $c\tau \approx 7.80$ m (the proper decay length of a pion, using its rest lifetime $\tau \approx 26.03$ ns)
- The probability of the pion decaying before reaching distance L is:

$$P_{\text{decay}} = 1 - e^{-L/\lambda} = 1 - e^{-L/(\beta\gamma c\tau)}$$

2. Decay Chances Over 100 cm ($L = 1.0$ m)

Using the two kinetic energy values:

1 GeV Kinetic Energy ($\beta\gamma \approx 8.16$)

- Mean decay length:

$$\lambda = 8.16 \times 7.80 \text{ m} \approx 63.6 \text{ m}$$

- Decay probability over 100 cm:

$$P_{\text{decay}} = 1 - e^{-1.0/63.6} = 1 - e^{-0.0157} \approx 1.56\%$$

- *(˜98.4% survival chance)*

2. Decay Chances Over 100 cm (Continued)

0.1 GeV Kinetic Energy ($\beta\gamma \approx 1.40$)

- Mean decay length:

$$\lambda = 1.40 \times 7.80 \text{ m} \approx 10.9 \text{ m}$$

- Decay probability over 100 cm:

$$P_{\text{decay}} = 1 - e^{-1.0/10.9} = 1 - e^{-0.0917} \approx 8.76\%$$

- *(˜91.2% survival chance)*

3. Decay Chances Over 1 cm ($L = 0.01$ m)

1 GeV Kinetic Energy ($\beta\gamma \approx 8.16$)

- Decay probability over 1 cm:

$$P_{\text{decay}} = 1 - e^{-0.01/63.6} = 1 - e^{-1.57\times10^{-4}} \approx 0.0157\%$$

- *(˜99.984% survival chance)*

0.1 GeV Kinetic Energy ($\beta\gamma \approx 1.40$)

- Decay probability over 1 cm:

$$P_{\text{decay}} = 1 - e^{-0.01/10.9} = 1 - e^{-9.17\times10^{-4}} \approx 0.0917\%$$

- *(˜99.908% survival chance)*

Impact on Tracking Efficiency

In a 100 cm tracking volume, approximately **8.8%** of 0.1 GeV pions will decay into a muon and a neutrino ($\pi^+ \rightarrow \mu^+ \nu_\mu$) before traversing the entire detector, causing a kinked or broken track, compared to only **1.6%** of 1 GeV pions.